UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE) IV-Semester (MAIN) Examinations Jun/Jul-2026

ENGINEERING MATHEMATICS – IV

(Probability & Statistics)

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	Define Bernoulli's distribution.	2	1	1
b)	Calculate mean of Gamma distribution.	2	3	2
c)	If X follows Poisson distribution with mean 2, find P(X=1).	2	3	
d)	Distinguish correlation and regression.	2	4	
e)	Define critical region and acceptance region.	2	1	
f)	Write normal equation for second degree polynomial.	2	2	
2. a)	A lot of 10 fuses contain 2 defectives fuses. A sample of 3 fuses is drawn at random without replacement. A random variable X denotes the number of defective fuses in the sample. Find the probability distribution X and it's a mean and variance.	6	4	
b)	If the median value and total frequency are 35 and 170 of an incomplete distribution is given below	6	3	

Variable	0-10	10-20	20-30	30-40	40-50	50-60	60-70
frequency	10	20	-	40	-	25	15

Find out missing frequency and calculate mean of the complete table.

3. a)	A mileage in thousands of kilometers which car owners get with a particular brand of tire is a random variable X having pdf $f(x) = \begin{cases} k\, e^{\frac{-x}{50}}, & for\ x > 0. \\ 0, & for\ x \leq 0. \end{cases}$ Find k value and the probability that a tire lasts (i) at most 10,000kms (ii) between 15,000 kms to 30,000 kms.	6		4

b) A university conducts a placement examination to all incoming freshmen. The examination scores of the 1000 examinees last semester were approximately normally distributed with a mean of 80 and a standard deviation of 5. Find the total number of examinees that randomly chosen student got scores below 70. Above 82? Between 75 and 90? $[P(0 \leq z \leq 0.4) = 0.1554, P(0 \leq z \leq 2) = 0.4772, P(0 \leq z \leq 1) = 0.3413)]$ 6 3

4. Explain Binomial distribution and find its mean and moment generating function. 6 3

b) Explain normal distribution and write its properties and applications. 6 2

5. a) Obtain the equation of the two lines of regression for the data given below 6 3

x	2	3	5	7	11	13
y	4.194	4.291	4.485	4.678	5.067	5.261

b) Calculate the rank correlation for the ages of husband and wife: 6 3

H	23	27	28	29	30	31	33	35	36	39
W	18	22	23	24	25	26	28	29	30	32

6. a) A sample survey of 800 families with 4 children has been made regarding the number of girls and boys in the families. The following data was obtained. 6 4

No. of families	64	236	290	178	32
No. of boys	4	3	2	1	0
No. of girls	0	1	2	3	4

Is the data consistent with the hypothesis that the male and female births are equally possible? Test at 5% level of significance.

$[\chi^2(0.05) = 9.488]$

b) In a large lot of electrical bulbs, the mean life and standard deviation of the bulbs are 360 hrs and 90 hrs, respectively. A sample of 625 bulbs was chosen. It is found that the mean life and standard deviation of the bulbs are 355 hrs and 90 hrs, respectively. Can we conclude that the sample is drawn from the given population? Test at the 5% level of significance and find the 95% confidence limits of the mean. (t =1.96) 6 4

7. a) Compute the coefficient of skewness and kurtosis based on the following data 6 3

x	4.5	14.5	24.5	34.5	44.5	54.5	64.5	74.5	84.5	94.5
f	1	5	12	22	17	9	4	3	1	1

b) Two random samples have the following values

Sample1	31	22	28	29	27	35
Sample2	27	21	25	23	26	

Test at 5% level of significance whether the two sample could have come from the same population.(F = 6.26)

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